

## Answers: Mean Class Size

Suppose a college is offering five sections of an introductory statistics course. The class sizes for the sections are:

| Section    | 301 | 302 | 303 | 304 | 305 |
|------------|-----|-----|-----|-----|-----|
| Class size | 10  | 4   | 5   | 3   | 6   |

a) Find the mean class size per section. What is the observational unit? What is the variable?

The observational unit is a **section** (there are 5). The variable is the **class size** of the section.

$$\bar{x} = \frac{10 + 4 + 5 + 3 + 6}{5} = \frac{28}{5} = 5.6 \text{ students}$$

b) Now ask all the students from all the sections what the class size of their section is. Find the mean class size per student. What is the observational unit? What is the variable?

The observational unit is a **student** (there are 28). The variable is the **class size of the student's section**, as the student reports it. Each student reports the class size of their own section:

| Section    | Students reporting | Value each reports | Total      |
|------------|--------------------|--------------------|------------|
| 301        | 10                 | 10                 | 100        |
| 302        | 4                  | 4                  | 16         |
| 303        | 5                  | 5                  | 25         |
| 304        | 3                  | 3                  | 9          |
| 305        | 6                  | 6                  | 36         |
| <b>All</b> | <b>28</b>          |                    | <b>186</b> |

$$\bar{x} = \frac{186}{28} \approx 6.64 \text{ students}$$

The mean is bigger from the students' point of view. More students are in the bigger sections (10 of the 28 are in section 301), so a typical student is in a bigger section than a typical section has. Which mean you get depends on what the observational units are.